MUHAMMAD AHMAD SOHAIL

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Personal Statement

Electrical and Electronic Engineering graduate from De Montfort University with a year of industry experience delivering production-ready PCB designs at a power electronics company, including a commercially completed 220 V AC to 12 V DC, 60 W power supply unit. Technical expertise spans isolated and non-isolated DC-DC conversion, three-phase SPWM motor drives, SiC MOSFET characterisation, thermal management, and multilayer PCB layout in Altium Designer, supported by hands-on simulation work in LTspice and Multisim. I am seeking a graduate role in power electronics or energy where I can contribute production-quality hardware solutions from day one.

Education

BEng (Hons) with Placement: **Electrical and Electronic Engineering (Expected 2:1) (2026)**
De Montfort University (Leicester).

Work Experience

Placement, 09/2024 – 08/2025
Automotive Devices

- **Power Electronics Design:** Successfully designed and implemented the PCB layout and schematic for a robust 220-240V AC to 12V DC, 60W power supply unit.
- The project was successfully completed within agreed timelines and met all key performance and design milestones.
- **IoT/Sensor System Development:** Engineered the complete schematic and PCB for the ESP32 Multi Sensor module.
- Designed the schematic and PCB for an RJ45-ESP32 module and worked on solutions involving the BlueNRG-M2 (Bluetooth Low Energy) module.
- Utilised Altium Designer to create high-quality schematics and multilayer PCB layouts, applying theoretical knowledge to real-world, innovative hardware solutions.

Projects

Individual Project: Supercapacitor Replacement for Li-ion Energy Storage

- Investigated replacing ENERGUS Li8p25RT (72 Wh) and Molicel INR-21700-P42A lithium-ion targets with Maxwell BCAP3400 (3400 F, 2.7 V) supercapacitor banks in a low-power energy harvesting application.
- Designed 2S×3P and 2S×14P series-parallel bank configurations through datasheet-driven energy analysis, equivalent capacitance derivation, and voltage swing calculations.
- Built and validated charge and discharge circuits in NI Multisim against a five-interval load profile (1.5 V standby to 5 V peak), achieving under 2% error across all intervals.
- Concluded BCAP3400 banks are technically viable where cycle life and peak power delivery are prioritised over gravimetric energy density.

Traffic Lights Controller with Concurrent Processing

- Developed a traffic light controller for a 2-way crossroads using ATMEGA328P in bare-metal C, implementing dynamic timing, priority modes, and real-time interrupt-driven responsiveness.
- Integrated accelerometer-based orientation detection for dynamic priority adjustment and ESP8266-based remote monitoring.

Multimeter and Function Generator

- Designed and built a multimeter and function generator with an ATMega328 microcontroller.
- Implemented DC (0-2V, 0-20V) and AC (0-20V peak-to-peak) voltage ranges with 10% accuracy.
- Developed a sinusoidal function generator (200Hz-7kHz, 5Vp-p).
- Constructed and tested on breadboard, transitioned to custom PCB with KiCad.
- Reduced circuit design errors by 10% through iterative testing and debugging using KiCad.

Rainwater Harvesting System Design and Management

- Led and collaborated with a team of five on a rainwater harvesting system to address water shortages in rural Cambodia.
- Led project planning and scheduling using Gantt charts and WBS.
- Conducted stakeholder analysis and ensured compliance with international standards.

Linear and Switched DC-DC Power Converter Analysis

- Designed and analysed shunt regulator, series-pass regulator, and buck converter circuits for comparative power electronics evaluation.
- Ran spreadsheet-based simulations and mathematical modelling to produce efficiency, line/load regulation, and temperature-dependence plots.
- Tested converter behaviour in laboratory sessions and documented experimental procedures, measurements, and results in a technical logbook.
- Compared circuit performance and identified suitable real-world applications based on efficiency, voltage stability, noise, and load conditions.

AC Induction Motor Drive for Low-Cost HEV E-Axle

- Developed an Arduino Mega-based control system for a prototype three-phase induction motor drive for a 48 VDC hybrid electric vehicle E-axle application.
- Generated PWM-controlled three-phase sinusoidal signals to operate a black-box power module and induction motor.
- Tested bridge switching in stages, from single-leg operation to three-phase waveform generation, with focus on speed and torque control.
- Maintained a technical logbook covering design decisions, hardware interfacing, testing results, and safety considerations.

Converters, Devices and Thermal Design

- Designed and analysed regulated linear DC-DC converters and compared performance against switched-mode alternatives across efficiency, noise, and load regulation.
- Designed isolated DC-DC converters with transformer turns ratio, gate drive timing, and component value specification from a given input/output power spec.
- Performed custom transformer design using ETD ferrite cores, specifying core geometry, winding arrangement, and flux density constraints.
- Evaluated power semiconductor devices (Silicon MOSFET, SiC MOSFET, IGBT) for blocking voltage, conduction losses.

Certifications

- Google Project Management Professional Certificate
- Nominated for Best Placement Student of the Year.

Professional Relevant Skills

- PCB Design (Altium, KiCad).
- Project Management
- LTspice / Multisim
- DIgSILENT PowerFactory (Basics)
- IPSA
- Digital Logic Design (Intel Quartus Prime lite)

Hobbies and Interests

Co-Founder of **ZamWheelz**, a vehicle trading platform developed for the Zambian market, where I lead an in-house software development and marketing team of 13 full-time employees. This role has strengthened my leadership, system architecture planning, product development, and cross-functional team management skills, while giving me direct experience in delivering and maintaining a live production platform.

I closely follow Formula 1, which reinforces my interest in high-performance engineering, embedded electronics, control systems, and real-time data-driven optimization.